Devireddy Lalith Sai

M.S. Researcher | Machine Learning | Computer Vision | Multimodal & Embodied AI

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RESEARCH INTERESTS

Machine learning, computer vision, and digital image processing across multimodal, embodied, and scientific applications of AI. Research interests include intelligent systems, simulation-based evaluation, vision-language-action policies, generalized zero-shot recognition, multimodal retrieval, optimal transport, and AI for materials science, with an emphasis on reliability, uncertainty quantification, and reproducible experimentation.

EDUCATION

M.S. in Artificial Intelligence Convergence Engineering

Mar 2025 - Feb 2027 (Expected)

Changwon National University (National Public University), Changwon, South Korea · **GPA: 4.11/4.50**

Intelligent Systems Lab (ISLab) · Advisor: Prof. Oh-Seol Kwon

M.S. Thesis: A Wavelet Neural Operator Action Head for Vision-Language-Action Models

B.Tech in Computer Science and Business Systems

Jul 2020 - May 2024

SRM Institute of Science and Technology, Chennai, India · **CGPA: 8.58/10** · QS Asia 2027: #226

Undergraduate Thesis: Pediatric Autism Spectrum Disorder Identification via Deep CNNs & Visual Explainability

SCHOLARSHIPS & ACHIEVEMENTS

- **3rd Place, Fish Species Recognition Challenge (FishONet), CV4Ecology/Codabench, 2026:** team lead; ranked among 103 participants on a 17,393-species generalized zero-shot benchmark; achieved 53.761% overall accuracy (77.00% photographed, 23.77% unseen).
- **BK21 FOUR Fellowship**, National Research Foundation of Korea, 3rd and 4th semesters, 2026.
- **Excellence Academic Scholarship**, Changwon National University, 3rd and 4th semesters, 2026.
- **Global Talent IELTS Scholarship**, Changwon National University, 1st semester, 2025.

RESEARCH EXPERIENCE

Graduate Research Assistant - Embodied AI & VLA Models

Mar 2025 - Present

Intelligent Systems Laboratory (ISLab), Changwon National University · Advisor: Prof. Oh-Seol Kwon

- **Thesis research on a Wavelet Neural Operator action head:** designed a single-pass 8.43M head, built on prior project work, replacing SmolVLA's 99.88M flow-matching expert: 11.8× smaller, 62× faster per chunk, 6× end-to-end. A pre-registered paired test of one seed on 8,000 new LIBERO placements gave 83.2% vs. 80.8% success; another seed had the expert ahead, a same-size regression head could not be distinguished from it, and 2 of 5 benchmarks passed non-inferiority.
- **Multi-GPU evaluation infrastructure:** established a reproducible baseline for the public SmolVLA checkpoint on LIBERO-Long (42.3% ± 2.1% success over 3 seeds and 600 episodes) and built multi-host evaluation across NVIDIA A100, Blackwell-class, and RTX A6000 systems.
- **Internal representation probing:** trained probes on frozen policy latent layers to detect the first object placement, achieving 0.876 held-out ROC-AUC (shuffled control 0.49) without fine-tuning backbone weights.
- **Failure analysis:** ran a one-seed pilot review of 53 failed rollouts, labeled by vision models (30% missed grasps, 28% stalled after the first subtask), and tested retry and stage-feedback modules under pre-registered gates. Both missed them: an oracle-triggered retry pilot gained 3.3 points against a 5-point bar, and stage feedback lost 11.7 points.
- **Research mentorship:** mentored four IIT (BHU) interns, defining research goals, experiment protocols, and codebase development across four separate VLA projects.

RESEARCH EXPERIENCE

Competitive ML Lead - CV4Ecology FishONet Benchmark **Jun 2026 - Sep 2026**

FishONet GitHub Repository

- Led the 3rd-place solution for a 17,393-species benchmark, combining closed-set classification, open-set retrieval, and a 12-feature logistic router.
- Ensembled three BioCLIP-2.5 backbones with retrieval and leak-free re-ranking; corrected validation leakage and obtained a confirmed +0.213 leaderboard gain.

Undergraduate Research Thesis - Autism Spectrum Disorder Identification **Jun 2023 - May 2024**

SRM Institute of Science and Technology · Department of Computer Science

- Trained VGG-16 and ResNet models for pediatric ASD classification, examined latent representations with t-SNE, and presented the explainability analysis orally at IRCCTSD'24.

RESEARCH FRAMEWORKS & SYSTEMS

SE3-VLA / GeoAct: SE(3) action head with geodesic loss, flow matching, and conformal uncertainty bounds · GitHub

SimRealDiag: comparison of simulated and physical trajectories across five kinematic metrics · GitHub

VLAProfiler: CUDA-event profiling across vision, language-model, fusion, and action-decoding stages · GitHub

VLA Research Survey: survey of 25+ models covering architectures, action spaces, and benchmarks · GitHub

PUBLICATIONS & TECHNICAL WRITING

- **D. Lalith Sai and O. S. Kwon.** "Design of a Lightweight Action Head for Vision-Language-Action Models." Korea Multimedia Society Fall Conference, Jeju, Nov. 2026. **Accepted.**
- **D. Lalith Sai and O. S. Kwon.** "Single-pass Action Chunk Generation for Vision-Language-Action Models Using a Wavelet Neural Operator." *Journal of Broadcast Engineering*. **Submitted.**
- "Machine Learning Approaches for Crystal System Classification in Lithium-Ion Battery Cathode Materials." *Batteries* (MDPI), 2026. **Co-author; under review.**
- Four-page paper on the WNO action head across five benchmarks, for an IEEE signal processing venue. **In preparation.**

CONFERENCES, EXHIBITIONS & PRESENTATIONS

- **Invited Exhibitor - 2026 ITRC Talent Development Expo**, COEX, Seoul (April 22-24, 2026): demonstrated embodied-AI research alongside the World IT Show.
- **Invited Exhibitor - 2025 ITRC Talent Development Expo**, COEX, Seoul, South Korea (April 24-26, 2025).
- **Oral Presentation - IRCCTSD'24**, Chennai, India (May 2024): "Autism Spectrum Disorder Identification Using Deep Learning."

TECHNICAL SKILLS & COMPETENCIES

Programming: Python (Expert), SQL, C/C++, MATLAB, Bash/Shell, LaTeX

Machine Learning & AI: PyTorch, Hugging Face, LeRobot, BioCLIP, open-set recognition, flow matching, neural operators, conformal prediction, CNNs, ViTs, LoRA, optimal transport

Embodied AI & Robotics: SmolVLA, OpenVLA, LIBERO, MuJoCo, robosuite, MetaWorld, SE(3) kinematics, trajectory optimization

Systems: Linux, multi-node GPU evaluation, CUDA, Docker, Git/GitHub CI/CD, Slurm, AWS

Tests & certifications: TOEIC 910 · IELTS 6.5 · Microsoft Azure AI Fundamentals **Languages:** English, Tamil, Telugu